COMPX221 Practical Test 2 2018A

First Name/Given Name	Username
Surname/Family Name	**ID Number**

Marking Scheme

Properties correctly defined in all classes	/6
Inheritance relationship is correct	/2
Constructors in all classes are correct	/3
Abstract `drawObject()` method in superclass and implementation in Plant and Feature classes is correct	/3
`growPlant()` method is correct	/2
`addPlant()` and `addFeature()` methods are correct	/2
`drawGarden()` method is correct	/1
`growGarden()` method is correct	/1
Total:	/20

Test Conditions

Practical Test 2 will be an open book test. You can refer to any printed notes, look at information on Moodle as well as use online help. You have **105** minutes to complete as much as you can.

You are not allowed to use any of the following software during the test:

- email applications
- file sharing applications
- facebook or other social media websites
- online chat applications

Any other applications which allows communication with other people.

You are not allowed to use USB drives during the test, so copy what you need into your account beforehand.

Any type of electronic device is not allowed, including laptops, tablets, phones, music players, etc.

Anyone found guilty of breaching these conditions will get sent to the University Discipline Committee.

Getting Started

Copy the **COMPX221-PracTest2-2018a** folder from the **COMPX221** library folder on L: drive. Go into the folder on your desktop and open the **PracTest2.pde** file in processing.

A screenshot is also provided to show you what your output should look like.

Task A – Create the AbstractObject class.

`AbstractObject` is an **abstract** class. It has the following **protected** fields:

- `position`: a PVector that stores the x and y of the centre of the object
- `objectColor`: the color of the object

It also has one abstract method called `drawObject` which doesn't have any parameters and doesn't return anything

The constructor will initialise the properties to the values x, y and colour values passed in to the constructor.

Task B – Create the Plant class.

The `Plant` class represents the different types of plants in the garden. It will inherit from the abstract class and it has the following extra fields:

- `size`: an integer which stores the width and height of the plant
- `name`: a string which stores the name of the plant

The `Plant` class will also have a private constant called `TEXT_SIZE` which is set to `12`. This constant is used to set the text size when the name is displayed in the sketch window.

The `Plant` class also has the following methods:

- A `constructor` that is passed the x, y, plant colour, size and name of the plant and initialises the object to those values.
- a `drawObject()` method that draws a filled in circle centred around the x and y position for the plant and displays the name of the plant centred inside the plant (use the `textAlign` method). Make sure you set the text size and set the fill colour for the text to black before drawing the text inside the plant. Draw the text after drawing the filled in circle.
- a `growPlant()` method that is passed the amount to grow and increases the size of the plant by the amount given.

After creating the class, test it by uncommenting the code in the sketch window to create and draw some plant objects. *Note: you must click in the sketch window first and then press a key to make the plants grow*.

Task C – Create the Feature class.

The `Feature` class is used to represent features in the garden like benches, water fountains, etc. It will inherit from the abstract class and it has the following extra fields:

- `width`: The width of the feature
- `length`: The length of the feature

The `Feature` class also has the following methods:

- A `constructor` that is passed the x, y, feature colour, width and height of the feature and initialises the object to those values.
- a `drawObject()` method that draws a filled in rectangle centred around the x and y position for the feature.

After creating the class, test it by uncommenting the code in the sketch window to create and draw some feature objects.

Task D – Create the Garden class.

The Garden class encapsulates everything about a garden. It has the following fields:

- `objectList`: Arraylist of objects in the garden

The `Garden` class also has the following methods:

- A `constructor` that creates a new arraylist object.
- An `addPlant()` method that is passed all information about a plant and will add a new plant to the garden.
- An `addFeature()` method that is passed all information about a feature and will add a new feature to the garden.
- A `drawGarden()` method that will draw all the objects in the garden.
- A `growGarden()` method that is passed the amount to grow the garden by and makes **only the plants** in the garden grow.

After creating the class, go to the sketch window code and comment out the testing code for the Plant and Feature objects. Then uncomment the code for the Garden object as well as the createGarden method. If you have any trouble uncommenting the code ask the demo for help.